Cross-Traffic Management System

System Requirements and Use Case Diagram

Team B6

System Requirements

Functional Requirements

FR-01 — The system shall detect vehicles arriving at the intersection from all four directions.

FR-02 — The system shall queue arriving vehicles in FIFO (First In, First Out) order.

FR-03 — The system shall manage intersection access using M/M/1 queuing (one vehicle at a time).

FR-04 — The system shall manage intersection access using M/M/2 queuing (two vehicles simultaneously on non-conflicting paths).

FR-05 — The system shall grant or deny crossing permission to each vehicle based on queue state.

FR-06 — The system shall detect potential collision risks before granting access.

FR-07 — The system shall prevent simultaneous crossing of conflicting vehicle paths.

FR-08 — The system shall support Vehicle-to-Vehicle (V2V) communication for position broadcasting.

FR-09 — The system shall support Vehicle-to-Infrastructure (V2I) communication for signal exchange.

FR-10 — The system shall monitor queue length continuously to detect and respond to congestion.

FR-11 — The system shall formally register a vehicle exit from the intersection zone to update queue state.

FR-12 — The system shall compare M/M/1 and M/M/2 scenarios to determine the optimal congestion reduction strategy.

Non-Functional Requirements

NFR-01 — The system shall respond to vehicle arrival and crossing events in real-time (deterministic timing guaranteed by freeRTOS).

NFR-02 — The system shall be deterministic, identical inputs must always produce identical outputs.

NFR-03 — The system shall handle a minimum of four simultaneous queues, one per direction.

NFR-04 — The system shall operate within the memory and processing constraints of an embedded hardware platform.

NFR-05 — The system implementation in C-code shall map directly and traceably to the UPPAAL formal models.

NFR-06 — The system shall maintain safe operation even under high vehicle arrival rates (traffic load stress).

NFR-07 — The V2V and V2I communication channels shall be assumed reliable for simulation purposes.

Use Case Diagram